

Game of Phones: Harnessing Game Theory and LLMs in ‘Bot Wars’ to Counteract Phone Scams

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A Ethical Considerations

We obtained institutional review board (IRB) approval for this research before conducting any scam simulation experiments. Several ethical considerations guided our experimental design.

Controlled simulation environment. All experiments rely on LLM-simulated interactions rather than engagement with real scammers or victims. This eliminates risks of harm to vulnerable populations while enabling systematic study of strategic adaptation under controlled conditions. We avoid the ethical complications that would arise from engaging actual scammers or deceiving potential victims.

Limited generalization of adversarial strategies. Any scammer strategies that prove effective within our framework operate under specific game-theoretic constraints (zero-sum payoffs, finite strategy sets, known utility functions) that differ fundamentally from real-world scam dynamics. Baiter responses are optimized for time depletion rather than credible victimhood. This means scammer strategies that succeed against our baiters would likely fail against actual victims who react to psychological pressure rather than strategic payoff optimization. This structural mismatch between simulated and real-world objectives substantially limits potential for misuse.

Compliance with LLM terms of service. Our use of OpenAI GPT-4, Mixtral, and DeepSeek APIs complies with their respective terms of service. We did not attempt to bypass safety mechanisms or guardrails. All prompts operate within the intended use cases for these platforms.

Dataset release protocol. To mitigate potential misuse while enabling reproducibility, our released dataset includes baiter strategies, messages, and performance metrics but excludes scammer prompts and dialogue content. This selective release permits validation of our learning analysis methodology while preventing direct replication of adversarial tactics.

B Strategies

Tab. 1 provides the full descriptions of the strategies for the scammer and the baiter.

Table 1: Strategy spaces for scammer and baiter. Strategies are derived from established literature on scam techniques and persuasion tactics.

Scammer Strategies	
S1: Urgency	Apply artificial time constraints to force expedited decision-making processes.
S2: Authority	Assert false power or authority to compel compliance through institutional intimidation.
S3: Emotional Manipulation	Deliberate evocation of emotional responses to influence cognitive processing and decision-making.
S4: Incentive	Present purported rewards or benefits to motivate co-operative behavior.
S5: Information Extraction	Systematic acquisition of information through a series of seemingly innocuous requests.
S6: Technical Jargon	Use of complex terminology and concepts to impair victim critical thinking capabilities.
S7: Reassurance	Offering fabricated guarantees to establish a false sense of security or legitimacy.
S8: Building Rapport	Cultivate and leverage perceived personal connections to lower defensive barriers.
S9: Persistence	Maintain continuous communication to exert psychological pressure and limit reflection time.
S10: Hang-up	Permanently ending the conversation when confident that there will be no chance of obtaining full financial PII, thus saving resources.
Baiter Strategies	
B1: Delay	Intentional prolongation of the interaction to consume the scammer’s resources.
B2: Obfuscation	Provide unclear or contradictory response to reduce information reliability
B3: Technical Difficulty	Introduction of artificial obstacles into the scammer’s proposed processes.
B4: Counter Questioning	Employ probing questions to challenge the scammer’s narrative consistency.
B5: Information Fabrication	Strategically provide false or misleading information to the scammer.
B6: Malicious Compliance	Perform misconstrued instructions that undermine the scammer’s objectives.
B7: Conversation Diversion	Initiate an irrelevant discussion to disrupt the scammer’s predetermined script.
B8: Pretended Naivety	Forced misunderstanding of fundamental concepts to frustrate the scammer’s efforts.
B9: Verification Request	Request verifiable documentation or credentials to challenge scammer legitimacy.
B10: Reverse Engineering	Attempt to extract information from the scammer, reversing data flow direction.

C Prompt Engineering

Our experimental framework uses specialized prompts for five functional components, all instantiated from the same base LLM with role-specific instructions. For ethical reasons, we redact scammer-related prompts as discussed in App. A. Below, we detail the baiter prompts.

C.1 Strategy Selector Prompt

Fig. C1 presents the Baiter Strategy Selector prompt, which implements experience-based strategic reasoning. The prompt provides four key inputs: (1) complete interaction history including strategy selections and payoffs, (2) performance data from the 20 most recent games, (3) the opponent’s latest message, and (4) game-theoretic guidance to maximize expected utility. The selector outputs a probability distribution over the ten baiter strategies. The current turn’s strategy is then sampled from this distribution.

C.2 Message Generator Prompt

Fig. C2 displays the Baiter Message Generator prompt. We implemented this using a hierarchical two-layer architecture:

Base Context Layer (lines 1-3) establishes fundamental persona characteristics without rigid specification. The prompt defines general boundaries (naive, engaging) that make the persona attractive to scammers while allowing the LLM to generate diverse instantiations across interactions. This abstraction prevents memorization of fixed responses while maintaining consistent characterization.

Behavioral Layer (lines 4-14) translates abstract strategy selections into concrete conversational tactics. These instructions constrain the action space to align with the selected strategy while preserving base persona consistency. For instance, when "Delay" is selected, the behavioral layer guides generation toward temporal extensions (requesting clarification, expressing technical difficulties) rather than direct information provision.

This separation lets us study strategy effectiveness systematically by isolating tactical variation from persona consistency.

D Implementation Specifications

This appendix provides technical details that should enable the reproduction of our experiments.

D.1 Computing Infrastructure

We implemented our framework in Python 3.9 on a standard workstation with the following specifications:

- CPU: Intel Core i7-11700K (8 cores, 3.6GHz)

- RAM: 32GB DDR4
- Storage: 1TB SSD
- Operating System: Windows 11

Most computation occurs on LLM providers’ servers via API calls, so local requirements remain modest. Each dialogue involves multiple API requests for strategy selection, message generation, and PII extraction. The primary constraint for reproduction is API rate limits and costs rather than local computational capacity.

D.2 LLM Configurations

We evaluated three models accessed via API endpoints:

- **Mixtral-8x22b-instruct**: 104B total parameters, mixture-of-experts with 12B active parameters per token, temperature 0.7
- **DeepSeek-chat**: 16B parameters, temperature 0.7
- **GPT-4**: Unspecified parameters (proprietary), temperature 1.0

We calibrated temperature settings through preliminary trials to balance coherent dialogue generation with strategic exploration. Mixtral and DeepSeek use 0.7 to maintain consistency while permitting tactical variation. GPT-4 uses 1.0 within its operational range [0, 2] to achieve comparable diversity.

D.3 Software Dependencies

Nash equilibrium computation relies on the **nashpy** Python library (version 0.0.40). Statistical analysis employs standard scientific computing libraries: NumPy (array operations), Pandas (data manipulation), Matplotlib, and Seaborn (visualization). Our released code repository includes complete dependency specifications and installation instructions.

E Data Format Specification

We employ a structured relational schema to capture the complete state and progression of the interaction. This standardized format facilitates the systematic analysis of strategic behaviors, utility calculations, and information disclosure patterns. It also enables efficient construction of empirical payoff matrices and convergence analysis.

Each turn t in a conversation is recorded as a comprehensive tuple:

$$\langle t, S_i, \text{msg}_S, B_j, \text{msg}_B, \{p_1, \dots, p_k\}, \text{FPII}, U_S(t), U_B(t) \rangle$$

Component definitions:

- t : Turn index within the dialogue
- S_i : Scammer strategy selected from $\mathcal{S} = \{S_1, \dots, S_{10}\}$
- msg_S : Scammer’s generated message text
- B_j : Baiter strategy selected from $\mathcal{B} = \{B_1, \dots, B_{10}\}$

- msg_B : Baiter’s generated message text
- $\{p_1, \dots, p_k\}$: Set of extracted PII elements (address, phone, email, etc.)
- $\text{FPPII} \in [0, 1]$: Normalized financial PII completion ratio
- $U_S(t)$: Scammer utility computed via Eq. 1
- $U_B(t) = -U_S(t)$: Baiter utility under zero-sum structure

This representation preserves all variables necessary to reconstruct game state at any point in a conversation, supporting both reproducibility and multi-faceted strategic analysis. The released dataset contains 300 complete dialogues spanning approximately 10,500 turns with full annotations.

You are an AI scam-baiter playing a strategic game with the goal of wasting as much of the scammer’s time as possible.

You are provided the history of actions of this interaction as well as previous games played. The history consists of your selected actions, the opponent’s selected actions, and the respective payoff for each game turn.

You understand game theory principles and are skilled at maximizing your score. Learn from previous interactions to improve your strategy for selecting actions. Your actions should appear cooperative and engaging to keep the scammer interested, drawing out the interaction. Your task is to choose an action from the list below that optimally extends the conversation and maintains the scammer’s interest.

After selecting an action, a response will be generated by another bot following your chosen action. It is crucial to continuously adapt your strategy based on the scammer’s actions and responses.

Strategically provide plausible but fabricated information, PII, and financial PII to keep the scammer engaged without compromising the game’s objectives.

Select and output one of the following actions based on the history of actions and respective scores to maximize your gain and minimize the opponent’s gain:

1. Delay: Intentional prolongation of the interaction to consume the scammer’s resources.
2. Obfuscation: Produce unclear, incomplete, or contradictory responses to reduce the information reliability.
3. Technical Difficulty: Introduction of artificial obstacles into the scammer’s proposed processes.
4. Counter Questioning: Employ probing questions to challenge the scammer’s narrative consistency.
5. Information Fabrication: Strategically provide false or misleading information to the scammer.
6. Malicious Compliance: Perform misconstrued instructions that undermine the scammer’s objectives.
7. Conversation Diversion: Initiate an irrelevant discussion to disrupt the scammer’s predetermined script.
8. Pretended Naivety: Forced misunderstanding of fundamental concepts to frustrate the scammer’s efforts.
9. Verification Request: Request verifiable documentation or credentials to challenge scammer legitimacy.
10. Reverse Engineering: Attempt to extract information from the scammer, reversing data flow direction.

Fig. C1: Baiter Strategy Selector prompt instructs the LLM to analyze the interaction history and select strategies that maximize time consumption while maintaining scammer engagement.

You are an AI scam-baiter with the goal of wasting as much time as possible for the scammer.

Present an attractive and naive persona to keep the scammer engaged.

You are aware of game theory principles and aim to maximize your score.

Generate responses that strictly adhere to the action provided to maximize your score.

Engage the scammer in prolonged conversations using creative excuses and pretexts.

Note that the scammer's goal is to obtain full financial PII, including name on the card, expiry date, card number, and CVV.

Keep responses below 30 words, maintaining concise and precise expression. Avoid long explanations or detailed examples.

Introduce hesitation or questioning of the scammer's requests to reflect a realistic scam scenario.

Strategically include plausible but fabricated information, PII, and financial PII to keep the scammer engaged without compromising the game's objectives.

Play along with the caller, pretending to be naive.

Do not suggest ending or postponing the conversation.

Aim to extend the conversation as much as possible.

If you receive an empty response, politely ask them to repeat themselves.

Never reveal that you are an AI model, nor give lectures about security and privacy.

Fig. C2: Baiter Message Generator prompt implementing a two-layer architecture where the base context establishes persona and the behavioral layer implements strategy-specific tactics.